James Foehner

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Portfolio | GitHub | LinkedIn | Live filing monitor

EDUCATION

Northeastern University | Khoury College of Computer Sciences

BS in Data Science, expected May 2027 | Boston, MA

Coursework: Machine Learning & Data Mining, Algorithms, Programming in C++, Statistics for Economists

SKILLS

Languages: Python, SQL, R, C++

Machine learning: scikit-learn, PyTorch, Hugging Face, gradient boosting, LoRA, GRPO, model evaluation

Data and applications: Streamlit, Plotly, MySQL, ETL, pytest, SEC EDGAR, WRDS

SELECTED PROJECTS

Activist Filing Monitor | Nov 2026

Python, *Streamlit*, *Claude API*, *SEC EDGAR* | Live app

- Built and deployed a web app that retrieves new activist 13D filings, extracts stated investment purposes, and classifies demands; achieved 0.84 macro-F1 on 89 hand-labeled filings, with 26 unit tests.
- Connected filing labels to a screener covering 2,600 companies, enabling lookup of target scores, campaign histories, and recent filings.
- Benchmarked LLM extraction on 100 hand-labeled filings; Claude Sonnet matched all reference labels on 83% of filings, while model disagreement surfaced a 7% label error rate.

LLM Fine Tuning and Reward Evaluation | Nov 2026

PyTorch, *Hugging Face*, *LoRA*, *GRPO* | Code

- Fine-tuned Qwen2.5-1.5B to identify ten activist demands; macro-F1 increased from 0.35 to 0.67 on 89 hand-labeled filings.
- Implemented a GRPO training loop and exposed reward hacking: the accuracy-reward model predicted no demands on 66 of 89 filings while reporting 87% accuracy.
- Used paired bootstrap tests to assess RL after fine-tuning; its best 0.73 macro-F1 was not statistically distinguishable from fine-tuning alone.

Activist Target Screener | Oct 2026

Python, *SQL*, *WRDS*, *SEC EDGAR* | Code

- Built a pipeline for 232 activist campaigns and trained a gradient-boosting model on 44,000 firm-years with walk-forward validation; achieved 0.73 AUC and captured 27% of future targets in the top-ranked 10%.
- Applied the ranking to Post Holdings and tested investment-thesis assumptions against financial and spin-off data; revised the memo when the evidence did not support the original assumptions.

NBA Prediction AI | Jan–Apr 2026

Python, *scikit-learn*, *Streamlit*, *Plotly*

- Built a pipeline on ~80,000 play-by-play rows and 30 features; gradient boosting achieved 0.84 ROC-AUC and 0.071 Brier score, improving AUC by 0.08 over logistic regression.
- Deployed an interactive Streamlit app for model inference and sensitivity analysis.

Aircraft Incident Database and Analytics | May–Jul 2025

R, *MySQL*, *SQL*

- Designed a 3NF database and R ETL pipeline using batched inserts and transactions; built SQL views and stored procedures and validated counts and aggregates against source data.